

Advanced Data Science Final Assessment

Radzim Sendyka, Christian Cabrera

University of Cambridge

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ADS Final Assessment

Available in Moodle:

Tuesday 19th of November 2024

Deadline for Practical 4 (check) in Moodle:

Tuesday 3rd of December 2024 at noon

Assessment check (Practical 4 session)

Tuesday 3rd of December 2024 at 3:00 pm

Deadline for final project report:

Friday 6th of December 2024 at noon

ADS Final Assessment

The maximum possible mark is 20:

Data Pipeline 1: 7 marks

- Predict the percentage of full-time students in a location in the UK.
 - The 2021 UK Census dataset
 - Open Street Maps (OSM)
- The prediction must use OSM features.
- Choose a second variable to predict using your pipeline.
- **Tell a story** about your data science process, design decisions, implementation and results. Your notebook should reflect the story. **The story is your report.**
- Structure your code using the fynesse framework.

ADS Final Assessment

The maximum possible mark is 20:

Data Pipeline 2: 13 marks: You have the **freedom to select the data problem** to address.

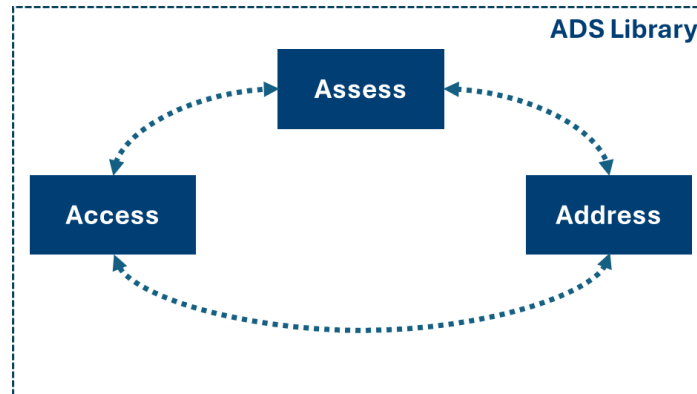
- We expect this pipeline to be more advanced than the first one. The first 7 marks will be allocated similarly to data pipeline 1. The remaining points are bonuses.
- Datasets you can use:
 - The 2021 UK Census dataset
 - Open Street Maps (OSM)
 - Price Paid dataset
 - Election dataset
- **Tell a story** about your data science process, design decisions, implementation and results. Your notebook should reflect the story. **The story is your report.**
- Structure your code using the fynesse framework.

Data Science

PURPOSE



Socio-Technical
Systems



Artificial
Intelligence

ADOPTION

Tell a Story

The motivation behind your data problems - **Problem First**

Consider the possible impact of your projects

- **Stakeholders of your system and different points of view of your system**

Document the design decisions you make and the different alternatives you explore

- **What did you try? What worked and what didn't? Why?**

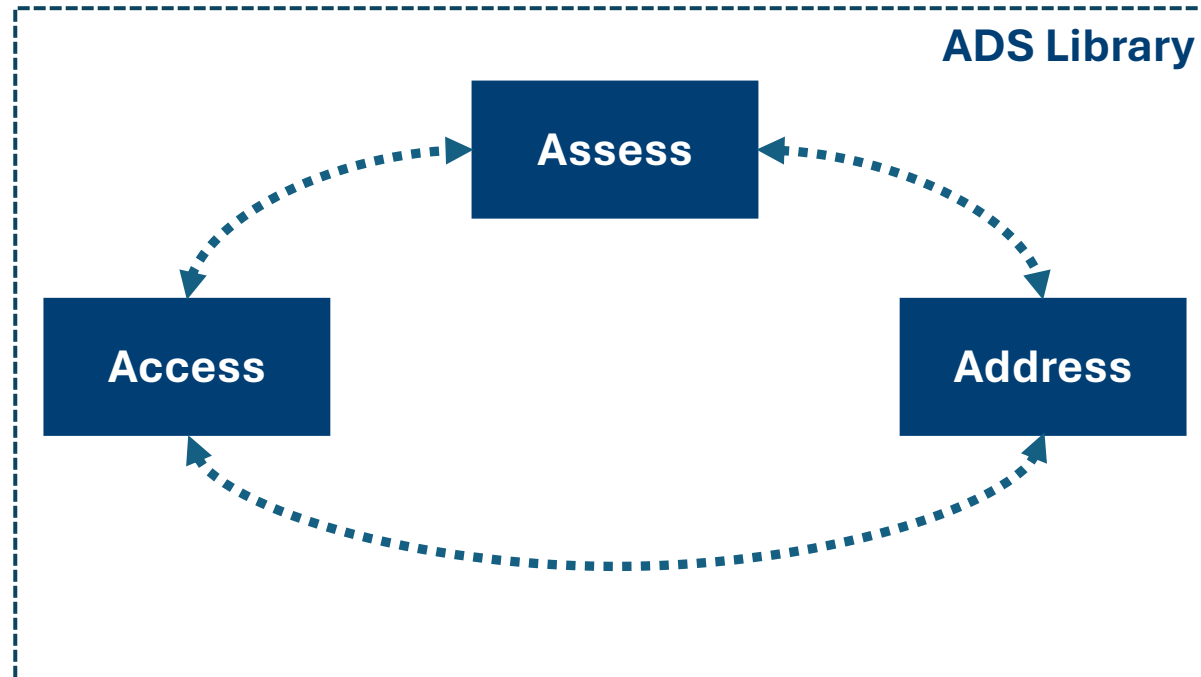
Document the system's architecture

- **Reusable and easy-to-extend system architecture that prioritises data (i.e., DOA)**

Document the process model you followed

- **Following a data science methodology (i.e., fynesse framework)**

Fynesse Framework



Access

Gaining access to data and making data available.

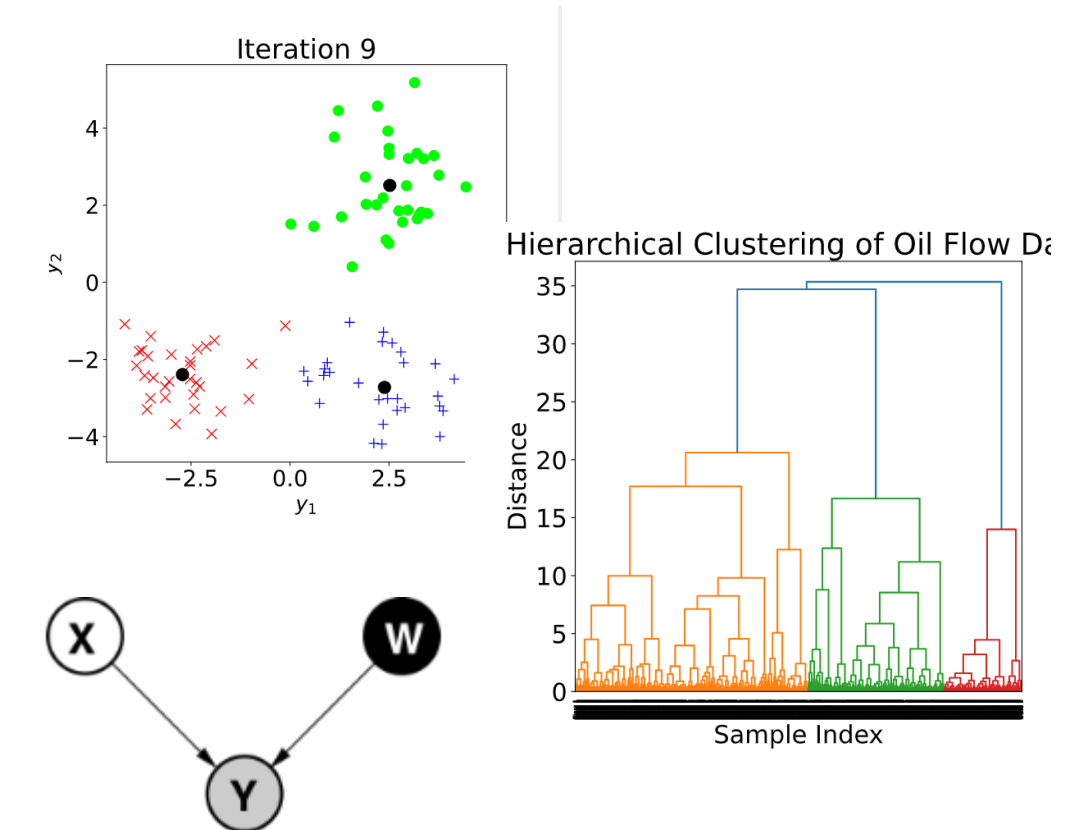
- **Practical 1**
 - Downloading and exploring data
 - Setting up a database
 - Upload data to the database
 - Joining datasets to enrich data
 - Online vs Offline joins



Assess

Understanding the nature of the data.

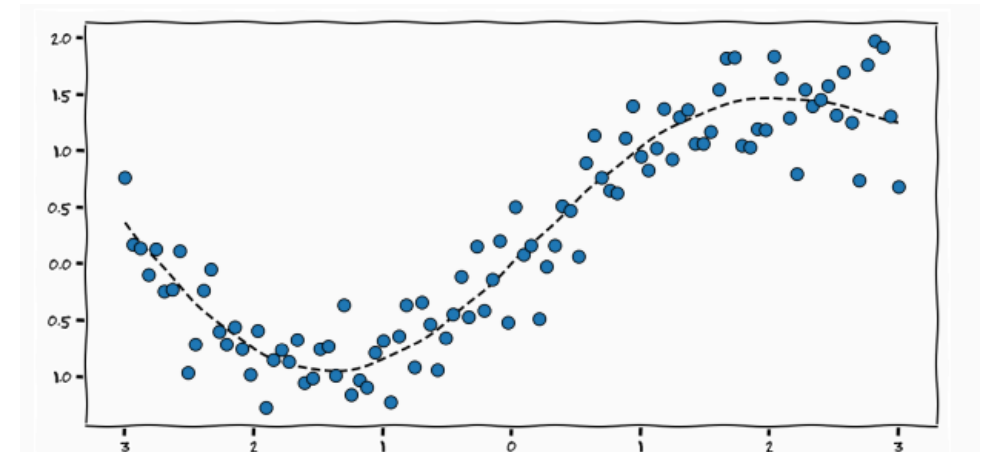
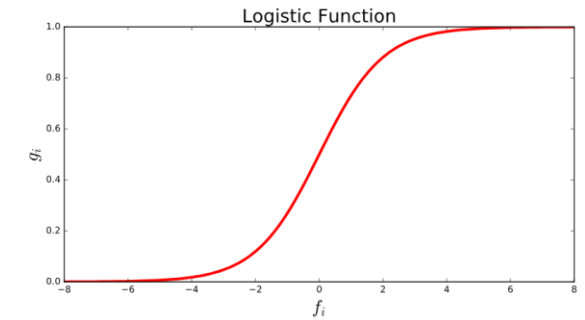
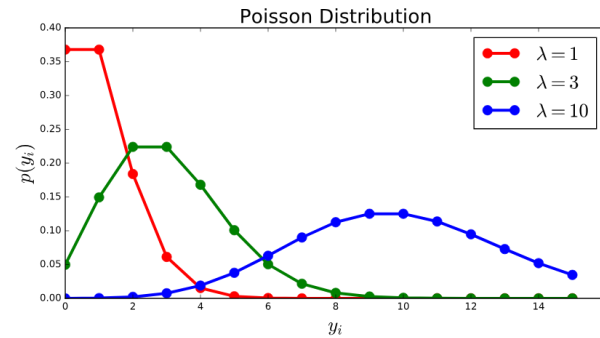
- **Practical 2**
 - Exploring the data
 - Missing values? Outliers?
 - How are variables represented?
 - Clustering
 - Dimensionality reduction
 - Principal Component Analysis



Address

Addressing the data problem.

- **Practical 3**
 - Linear regression model
 - Select the model
 - Try different model parameters
 - Compare models' performance
 - Select the best one
 - Can you automate this process?



Final Report

The final version of your notebook with the points discussed before

The GitHub link to your Fynesse library. Your repository should be public for now

A working database

Many thanks!